

JEE MAIN REVISION MASTER: THERMODYNAMICS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Enthalpy, Hess's Law & Bond Energies	8 PYQs	Thermochemistry
Moderate	Entropy & Gibbs Free Energy Spontaneity	6 PYQs	Spontaneity

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Enthalpy, Hess's Law & Bond Energies	$\Delta H = \Delta U + \Delta n_g RT, \quad \Delta_r H^\circ = \sum \Delta_f H^\circ(\text{products}) - \sum \Delta_f H^\circ(\text{reactants})$
Entropy & Gibbs Free Energy Spontaneity	$\Delta G = \Delta H - T\Delta S, \quad \Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0, \quad \Delta G^\circ = -RT \ln K$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Enthalpy, Hess's Law & Bond Energies (High Yield — 8 PYQs)

Core Principle: State functions, standard enthalpy of formation, Hess's law of constant heat summation, and bond energy calculations.

■ **Caution / JEE Pitfall:** Standard enthalpy of formation $\Delta_f H^\circ$ of pure elements in their standard reference state is strictly ZERO (e.g. $O_2(g)$, $C_{\text{(graphite)}}$).

■ **Speed Trick:** When using bond enthalpies: $\Delta H_{\text{(reaction)}} = \sum \text{BE}(\text{bonds broken in reactants}) - \sum \text{BE}(\text{bonds formed in products})$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A gas can be taken from A to B via two different paths ACB and ADB. When path ACB is used, 60 J of heat flows into the system and 30 J o..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Which one of the following reactions has $\Delta H = \Delta U$?..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Identify the incorrect statements among the following: (a) All enthalpies of fusion are positive (b) The magnitude of enthalpy change does n..." [Ans: N/A]

3.2 Entropy & Gibbs Free Energy Spontaneity (Moderate Yield — 6 PYQs)

Core Principle: Second and third laws of thermodynamics, entropy changes in phase transitions, and Gibbs criteria for spontaneous change.

■ **Caution / JEE Pitfall:** For an endothermic reaction ($\Delta H > 0$) with $\Delta S > 0$, the reaction is spontaneous ONLY at HIGH temperatures where $T > (\Delta H / \Delta S)$.

■ **Speed Trick:** At equilibrium: $\Delta G = 0$ implies $T_{\text{(equilibrium)}} = (\Delta H / \Delta S)$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Which of the following is a correct statement for a thermodynamic system?..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "A: Entropy of a perfect crystalline solid at absolute zero approaches zero. B: For spontaneity of a reaction, $T\Delta S > \Delta H$. Among t..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "Given below are two statements. Statement-I : Adiabatic work done is positive when work is done on the system and internal energy of the sys..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A gas can be taken from A to B via two different paths ACB and ADB. When path ACB is used, 60 J of heat flows into the system and 30 J of work is done by the system. If path ADB is used, work done by the system is 10 J. The heat flow into the system in path ADB is

- (A) 80 J
- (B) 20 J
- (C) 100 J
- (D) 40 J

Correct Answer: None

Solution Analysis: By the first law of thermodynamics, $\Delta U = Q - W$. Here, Q is the heat supplied to the system, and W is the work done by the system. For path ACB: Heat added, $Q_{\text{(ACB)}} = 60$ J Work done, $W_{\text{(ACB)}} = 30$ J So, $\Delta U = 60 - 30 = 30$ J The change in internal energy depends only on the initial state A and final state B. It does not depend on the path followed. So, for path ADB also, $\Delta U = 30$ J For path ADB: Work done, $W_{\text{(ADB)}} = 10$ J Using the first law again, $\Delta U = Q_{\text{(ADB)}} - W_{\text{(ADB)}}$ $30 = Q_{\text{(ADB)}} - 10$ $Q_{\text{(ADB)}} = 30 + 10 = 40$ J The heat flow into the system in path ADB is 40 J.

Benchmark Problem 2: KCET 2026 (Q#2)

Which of the following is a correct statement for a thermodynamic system?

- (A) The internal energy changes in all processes
- (B) Internal energy and entropy are state functions
- (C) Work is a state function
- (D) The work done in an adiabatic process is always zero

Correct Answer: None

Solution Analysis: The correct statement is: Option B: Internal energy and entropy are state functions. In thermodynamics, a state function is a quantity whose value depends only on the present state of the system, not on the path followed to reach that state. Internal energy U depends only on the state of the system, such as temperature, pressure, and volume. So it is a state function. Entropy S also depends only on the state of the system. So entropy is also a state function. So, internal energy and entropy are state functions. Now, the other options are not correct: Option A says internal energy changes in all processes, which is not true. Option C says work is a state function, which is not true. Option D says the work done in an adiabatic process is always zero, which is not true.